

MME8215: Operations Research - Seminar

I. Linear Programming

Linear Programming (LP) is a classic technique in Operations Research and was invented independently several times during World War II. With the invention of the Simplex algorithm, by George Dantzig (1947), the models could also be optimized, i.e. the best solution values of the decision variables given an objective and a set of constraints.

Let us see some LP models: "Incredible Chairs" and "Incredible Chairs II".

1.1 Incredible Chairs

The company Incredible Chairs produces two different types of chairs, A and B, where one unit (pallet of chairs) of Chair A can be sold for a profit of 4 and one unit of Chair B can be sold for a profit of 6. The chairs are produced on three production lines:

1. Line 1 requires 2 hours to produce one unit of Chair A, and at most 14 hours can be used due to other production on the line.
2. Line 2 requires 3 hours to produce one unit of Chair B, and at most 15 hours can be used.
3. Line 3 requires 4 hours to produce one unit of Chair A and 3 hours to produce one unit of chair B, and at most 36 hours can be used.

Suppose that you are the production planner at the company, how will you plan the production of the chairs? The first important thing to identify is what you are in control of. In other words, what are the decisions you can make as a production planner? In this case, you can decide **how many** of each type of chair to produce. Since these quantities are unknown and must be decided, we define two decision variables, $x_A \geq 0$ and $x_B \geq 0$, which will store the quantity of each type of chair. Notice that the variables have a domain, i.e. an interval of feasible values. Here, they are each constrained to take a positive value. The job of the production planner is to find the best possible values for these two variables.

There are many possible solutions represented by the decision variables (x_A and x_B). As the production planner, you are, not interested in a random production plan, stated in the variables x_A and x_B , but you want the **best feasible** production plan. To compare the different solutions, an **objective function** $f(x_A, x_B)$ which quantifies the utility of the solution is used. For this example, the objective function measures the profit of the solution obtained by producing x_A and x_B chairs. The objective function can therefore be stated as:

$$\max f(x_A, x_B) = 4 \cdot x_A + 6 \cdot x_B.$$

Not all solutions are possible; they have to be **feasible**. This means that the values assigned to the variables should satisfy the restrictions that are imposed by the production lines. In particular, for each production line, the planned production of chairs cannot exceed the available resource limit. On line 3, the sum of the required resources cannot exceed 36. Given the unknown values of the decision variables, we can now formulate an equation - a **constraint** - that limits the values of x_A and x_B to values that are possible given the limitations on line 3:

$$4 \cdot x_A + 3 \cdot x_B \leq 36.$$

Similarly, there are two other constraints imposed by lines 1 and 2. They are given below:

$$2 \cdot x_A \leq 14$$

$$3 \cdot x_B \leq 15.$$

The complete LP model is therefore:

$$\begin{array}{ll} \text{Maximize} & 4 \cdot x_A + 6 \cdot x_B \\ \text{subject to} & \\ & 2 \cdot x_A \leq 14, \\ & 3 \cdot x_B \leq 15, \\ & 4 \cdot x_A + 3 \cdot x_B \leq 36, \\ & x_A \geq 0, \\ & x_B \geq 0. \end{array}$$

LP models are linear, i.e., the only allowed calculations in objective function and constraints are multiplications between constants and decision variables, and the addition and subtraction of variables.

Solve the problem.

1.2 Incredible Chairs 2

What if the company were not producing two different chairs on three production lines, but say ten different chairs on five different production lines? If the same approach were used, the size of the LP model would increase, and it would be quite tedious to input using the direct approach (1.1). In this case, we can formulate a more abstract model. Given a set of chairs C , indexed by c , and a set of production lines P , indexed by p , we can define a (one-dimensional) vector $Profit_c$, which states the profit for each chair $c \in C$, a (one-dimensional) production capacity vector $Capacity_p$, which states the production capacity of each production line $p \in P$, and a (two-dimensional) matrix $RecourseUsage_{p,c}$ that states how much of the available resource on production line p is needed for production of one unite of chair c .

This gives the following LP model:

$$\begin{array}{ll} \text{Maximize} & \sum_{c \in C} Profit_c \cdot x_c \\ \text{subject to} & \\ & \sum_{c \in C} RecourseUsage_{p,c} \cdot x_c \leq Capacity_p, \forall p \in P, \\ & x_c \geq 0, \forall c \in C. \end{array}$$

The above model is a very generic (chair) production model for profit maximization for any number of chairs and any number of production lines. In fact, any (simple) profit maximizing production problem with a number of products and production resources can be modeled in this way.

Write code that solves the generic problem.

1.3 Diet problem

We wish to cook a nutritionally balanced meal by choosing the quantity of each food f to eat from a set of foods F in our kitchen.

Each food f has a cost, c_f , as well as a macro-nutrient profile $a_{m,f}$ for each macro-nutrient $m \in M$.

Because we care about a nutritionally balanced meal, we set some minimum and maximum limits for each nutrient, which we denote l_m and u_m respectively.

Furthermore, because we are optimizers, we seek the minimum cost solution.

With a little effort, we can formulate our dinner problem as the following linear program:

$$\begin{aligned} & \text{Minimize } \sum_{f \in F} c_f x_f \\ & \text{subject to } l_m \leq \sum_{f \in F} a_{m,f} x_f \leq u_m, \quad \forall m \in M \\ & \quad \quad \quad x_f \geq 0, \quad \quad \quad \forall f \in F. \end{aligned}$$

Solve the diet problem.

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II. Linear Programming cont.

Propose a model for solving the following problems. When applicable solve the problem graphically and algorithmically (try to draw the feasible solution space and solve the problem with simplex and/or using your implementation or an appropriate API).

1. Production Mix Production Mix A company produces two kinds of products, $P1$ and $P2$. The selling price of $P1$ is 300€ The selling price of $P2$ is 250€

- $P1$ requires 2 operator man-hours and 1 hour of machine operating time.
- $P2$ requires 1 operator man-hours and 3 hours of machine time.
- Operator man-hours cannot be higher than 40 hours per day.
- Machine operating time cannot be higher than 45 hours per day.
- Marketing states that $P1$ units per day cannot be higher than 12.

How many units of $P1$ and $P2$ should the company produce every day to maximise income?

2. Ski Patrol Ski Patrol Ltd. manufactures two different models of snowboards branded Normal and Deluxe with a weekly production plan and inventory. They sell their material to local shops. The company has a profit of 300€ per Normal snowboard and 400€ per Deluxe snowboard. The manufacturing process consists of roughing and polishing.

The Normal snowboard model requires 20 minutes of roughing and $\frac{2}{3}$ minutes of polishing. The Deluxe snowboard model requires 40 minutes of roughing and 1 minute of polishing. The roughing equipment is available for 1000 minutes per week and the polishing equipment for 800 minutes per week. Ski Patrol has already signed a contract with a local shop called Downhill Racer to deliver 18 Normal snowboards per week.

Ski Patrol hires you to formulate a Linear Programming Problem to help them determine the best production mix to maximise profits.

3. The good carpenter “The Fog Ltd” is a modern furniture company that makes tables and chairs. The selling price of a table is 30€ and the selling price of chairs is 10€. The carpenter at the company works up to 40 hours and needs 6 hours to make a table and 3 hours to make a chair. Customer demand requires that he makes at least 3 times as many chairs as tables. Tables take up four times as much storage space as chairs and there is room for at most four tables each week.

What is the optimal weekly production plan that optimises revenues? Formulate this problem as a linear programming problem.

4. Blending Problem The Kahuna company manufactures sausages using three kinds of meat. The relevant information about the ingredients is provided in the table below:

Ingredient	Cost (€/kg)	Availability (kg)
Pork	4.32	30€
Wheat	2.46	20€
Starch	1.86	17€

The company makes two types of sausages:

- Economy ($\geq 40\%$ Pork)
- Premium ($\geq 60\%$ Pork)

One sausage is 50 grams (0.05 kg). According to government regulations, the most starch that can be used in sausages is 25%. The company has a contract with a butcher, and have already purchased 23 kg pork, that will go bad if it's not used. The company has a demand for 350 economy sausages and 500 premium sausages.

Write a linear program to figure out the most cost-effective recipe to manufacture our sausages.

5. Petroleum Blending Harry Stamper is an engineer working on a petroleum company. The company produces three grades of motor oil – *super*, *premium*, and *extra*. The three grades are made from the same 3 ingredients, named component 1, component 2 and component 3. The company wants to determine the optimal mix of the 3 components in each grade of motor oil that will maximize profit.

The maximum availability of each component and their cost per barrel are as follows: To ensure

Component	Max barrels available / day	Cost / barrel
1	4500	12€
2	2700	10€
3	3500	14€

the appropriate blend, each grade must have a minimum amount of component 1 plus a combination of other components as follows:

Grade	Component 1	Component 2	Component 3	Selling price/barrel
Super	$\geq 50\%$	$\leq 30\%$	-	23€
Premium	$\geq 40\%$	-	$\leq 25\%$	20€
Extra	$\geq 60\%$	$\geq 10\%$	-	18€

The company wants to make at least 3000 barrels of each blend. Formulate an LP to help Harry Stamper find the optimal blend that maximises profit.

Analysis questions (Do this part once you have reviewed LP analysis including sensitivity analysis questions)

- What is the total profit obtained by the company with the Extra grade with the solution provided by the solver?
- What is the maximum value that the profit per barrel of component 2, in the Extra grade, can take without changing the base of the optimal solution?

- If the cost of the barrel of component 3 is increased by 1€, will the optimal composition change? Motivate your response
- Do we use every barrel available of every component in the optimal solution? Or do we have any surplus availability for any component? Motivate your response
- The provider of Component 2 may have a problem with the delivery and the availability may be reduced to 400 barrels. Will this shortage affect the optimal composition? Motivate your response
- Grace wants to use the results to renegotiate the terms and conditions with the suppliers of the different components. Which component is a better candidate to increase the availability? How would you use the model to negotiate new terms with the provider, assuming you do not want any change in the base of the optimal solution? Motivate your response

6. All gold canyon A gold seeker needs to find at least 2 kg of bronze and 3 kg of copper to pay his debts. There are two rivers in which the miner can find these precious metals. Each day the miner spends in river *A*, he finds 0.3 kg of bronze and 0.3 kg of copper while each day he spends in river *B*, he finds 0.2 kg of bronze and 0.4 kg of copper.

Can you help this gold seeker to pay his debts as soon as possible?

7. Making weapons Din Djarin is a retired bounty hunter that now runs a small factory specialised in making classic weapons in a colony recently established on Mandalore. The factory manufactures two different types of weapons, swords and spears. Din uses special alloys, mixtures of metals and rare materials, which have made a name for the endurance of his weapons across the colony. The table below shows the weight in pounds of each material needed to make each type of weapon:

Metal	Spear	Sword
Beskar	1	0.5€
Tungsten	1	1€
Titanium	3	1€

The total amount of each metal available, also in pounds, is: Beskar - 125, Tungsten - 225, Titanium - 300. The price of a spear is 40 credits and the price of a sword is 5 credits. Din has already sold 100 swords, so he needs to manufacture at least that many.

Write a LP model to find the optimal number of spears and swords that maximise revenue.

8. Chappie Chappie Ltd. has 3 production lines (A, B, and C) to make 4 different types of metal sheets of width 0.4, 0.5, 0.6, and 0.7 mm. The company needs to meet the monthly demand for each kind of product (in tons) expressed in the next table:

Sheet 0.4 mm	Sheet 0.5 mm	Sheet 0.6 mm	Sheet 0.7 mm
500	1200	1500	300

The production lines do not have the same overall efficiency for all types of sheets. Furthermore, due to design constraints, not all the lines are capable of manufacturing all types of sheets. The table below indicates the manufacturing costs in € of the different products in the different lines: a dash “-” means that it is not possible to manufacture a product in a line:

Line	Sheet 0.4 mm	Sheet 0.5 mm	Sheet 0.6 mm	Sheet 0.7 mm
A	60	50	50	45
B	80	70	75	70
C	-	60	60	-

The different lines have the following capacities (in hours) in the planning period: A - 500, B - 480, C - 370.

The following table represents the manufacturing time (hours) per ton of product required for each type of metal sheet in the different lines: Formulate a linear programming problem to obtain

Line	Sheet 0.4 mm	Sheet 0.5 mm	Sheet 0.6 mm	Sheet 0.7 mm
A	0.4	0.3	0.3	0.25
B	0.80	0.6	0.75	0.6
C	-	0.5	0.4	-

the monthly production plan (quantity of metal sheet in tons of each type of metal sheet to be manufactured in each production line).

9. Soft drinks Spellman's Ltd is a company that produces two types of chilling soft drinks: A , and B . Both beverages use a semi-elaborate C , another expensive ingredient D and other ingredients that are not relevant for production planning.

The selling price of drink A is 3€/liter and the selling price of drink B is 2€/liter.

1 liter of drink A uses 3 grams of ingredient D . A liter of drink B uses 1 gram of ingredient D . There are only 3 grams of ingredient D available per day.

The factory only has one mixer to elaborate both drink types and the semi-elaborate. It takes 1 hour to process a liter of drink A , 1 hour to process 1 liter of drink B , and 1 hour to process 1 cl of semi-elaborate C . The mixer is available 6 hours per day.

Drink A uses 2cl of semi-elaborate C and drink B uses 1cl of semi-elaborate C . The company has 3cl of semi-elaborate C plus the amount they decide to produce every day.

The following problem model provides the optimal daily production:

Decision variables:

- x_A : Production of drink A in liters,
- x_B : Production of drink B in liters,
- x_C : Production of semi-elaborate C in centiliters.

$x_A, x_B, x_C \in \mathbb{R}$.

Objective function:

$$\max z = 3 \cdot x_A + 2 \cdot x_B$$

z is the profit in euros.

Constraints

Availability of ingredient D in grams:

$$3 \cdot x_A + x_B \leq 3.$$

Availability of mixer in hours:

$$x_A + x_B + x_C \leq 6.$$

Availability of semi-elaborate C in centiliters:

$$2 \cdot x_A + x_B - x_C \leq 3.$$

Logical constraint: $x_A, x_C, x_B \geq 0$.

A solver provides the following solution:

Optimal

Total profit is 6.00 €

The following table shows the decision variables:

j	Variables	Solution	Reduced cost	Objective Coefficient	Objective Lower bound	Objective Upper bound
A	Production A (Liters)	0	-3	3	-inf	6
B	Production B (Liters)	3	0	2	1	inf
C	Production C (Liters)	3	0	0	-0	1.5

The following table shows the constraints:

j	Constraint	Slack	Shadow Price	Right Hand Side	Min RHS	Max RHS
0	Availability of ingredient D (Liters)	0	2	3	0	4.5
1	Availability of mixer (hours)	0	0	6	3	inf
2	Availability of semi-elaborate (Liters)	3	0	3	0	inf

Answer the following analysis questions:

1. Operational problems in the mixer, like mechanical problems, could reduce the available mixer. Based on the provided solution, how is the company able to deal with this type of problem? Motivate your response.
2. The managers of the company want you to negotiate with a new supplier that could increase your daily supply of ingredient D. What would be your terms for negotiating with the new supplier, ensuring that the basic variables remain the same?
3. The managers of the company would like to review the prices of both products. Analyse how changes in the prices of both products could affect the production plan and provide useful feedback to guide the revision of the prices.

Duality

Write down the dual problem.

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III. Transportation models.

Problem set

1. Distribution of pharma The US Department of Energy conducts a series of experiments in two research facilities located in Hawkins (Atlanta) and Los Angeles. The department needs to supply these facilities with pharma products that are processed in two factories in Portland and Flint. The delivery costs are summarised in the following table:

Production plants	Research Facilities	Delivery costs
Portland	Los Angeles	€30
Portland	Hawkins	€40
Flint	Los Angeles	€60
Flint	Hawkins	€50

The supply and demand data is provided in the table below:

Supply	Available units	Demand	Required units
Portland	200	Los Angeles	300
Flint	600	Hawkins	400

Determines how to supply the two research facilities at minimum cost.

2. Transport Planning A firm must transport machines from production plants A, B and C to warehouses X, Y and Z. Five machines are required in X, 4 in Y and 3 in Z, whereas 8 machines are available in A, 5 in B and 3 in C. The transport costs (in euros) between sites are provided in the table below.

Plant/Warehouse	X	Y	Z
A	50	60	30
B	60	40	20
C	40	70	30

1. Formulate an integer linear programming model that minimizes transport costs.
2. Assume that the cost of transporting a machine from plant B increases by €10 for all the machines as of the third one; that is, the 4th, the 5th, etc. Reformulate the model in Section a) by considering this assumption.

3. Planning production in a plastic injection company Patrick “Eel” O’Brian’s (PEOB) Ltd. is a company that manufactures plastic components for the automotive sector. As an intern at PEOB Ltd, your first assignment is to write an Integer Programming Model to calculate the optimal production plan for the plastic injection machine. The injection machine heats plastic and injects it into a mould to get a specific shape. Each mould can thus only be used to manufacture specific component types. Your first version will take into account the storage costs and the delayed orders cost which are defined below.

- *Storage costs* The storage or inventory costs represent the cost of storing the inventory levels of every component type at every planning period and are modeled as a fixed cost per unit and planning period. The storage costs are different for every component type.
- *Delayed orders costs* At every planning period, you may delay part of the demand for the next planning period. Customers will apply a fixed penalty for every delayed unit at every period. The delayed costs are also different for every component type.

Your model needs to take into account the following additional data:

- *Initial Inventory Levels (units)*: There is an initial inventory level for every component type available at the first planning period that needs to be taken into account.
- *Minimum and Maximum Inventory Levels (units)*: There is a maximum and a minimum inventory level for every component type.
- *Machine capacity (units)*: The machine capacity represents the number of units of a given component type that the machine can produce using a given mould. If a mould cannot be used to manufacture a component type, the machine capacity is zero for that combination of component type and mould.
- *Demand (Units)*: The company has several confirmed orders for the following periods and therefore, our model needs to take into account the demand for every product at every planning period.

- a. Write down the indexes and decision variables.
- b. Write down the objective function
- c. Write down the constraints

You have successfully validated your model, and it is already providing valuable information for the company. Now the company would like to extend the model to take into account as well the set-up costs:

- *Set up costs* As explained above, the injection machine uses different moulds to manufacture different parts. Each time that a mould is changed to make a different component, the operators need to set up the machine. The company estimates that this setup cost only depends on the mould that is used and is only applied when a product is changed.
- d. Modify the model to take into account the set-up costs.

4. Production Planning PLACASA, a supplier of plastic injection parts for the automotive sector, foresees the demand of an article for the following 4 months of 100, 140, 210 and 180 units, respectively. The firm can maintain only sufficient stock to meet the demand of each month, or it can have excess stock to meet the demand of two successive months or more with a storage cost of 1.2€ per month and unit of excessive stock. PLACASA calculates that the production cost for the following months, is €15, €12, €10 and €14 per unit. A preparation cost of €200 is incurred every time a production order is placed. The firm wishes to develop a production plan that minimises the total costs of production orders, of production and of keeping an article in stock.

Formulate and solve the shortest path model to find an optimum production plan.

5. Production Sequencing A manufacturer of the metal sector in Stafford manufactures four types of products in sequence on two machines. The Table below provides the necessary technical production details. Taking into account only the data on the production times per product and the maximum daily production capacity of each machine, and taking a maximum daily production of 100 units of each product type, consider and solve a maximal flow model in a graph to determine the maximum number of minutes during which machines 1 and 2 can operate on a production-in-sequence basis given the existing capacity constraints.

Machine	Product 1	Product 2	Product 3	Product 4	Daily Production Capacity (minutes)
1	2	3	4	2	500
2	3	2	1	2	380

6. Spellman's Distribution Network Optimization Spellman's would like Sabrina to focus on the optimization of their distribution network, to support the distribution of the 2 types of drinks from 3 sourcing facilities to 4 destination warehouses, through 6 intermediate warehouses. Sabrina needs to take into account the transportation costs (both from sourcing warehouses to intermediate warehouses and from intermediate warehouses to final destinations). Additionally, Sabrina can decide how many of the 6 intermediate solutions to use, given that they have a fixed operational cost that needs to be paid when the warehouse is used.

The following tables contain the transportation problem data:

Capacities of Sourcing Facilities (in units)

Sourcing Warehouse	Product A Capacity (units)	Product B Capacity (units)
S1	300	400
S2	250	150
S3	350	250

Note that the distribution costs are independent of the product type.

1. Help Sabrina write an Integer Programming Problem to model the transportation problem .
2. How would a Greedy algorithm solve the problem? Motivate your response.

Demands at Final Destinations (in units)

Destination Warehouse	Product A Demand (units)	Product B Demand (units)
D1	150	150
D2	200	250
D3	175	175
D4	225	175

Sourcing to Intermediate Warehouses Transportation Costs (Euros)

From/To	I1	I2	I3	I4	I5	I6
S1	2	4	5	7	3	6
S2	3	2	6	8	4	5
S3	1	3	4	6	5	7

Intermediate to Destination Warehouses Transportation Costs (Euros)

From/To	D1	D2	D3	D4
I1	5	2	3	4
I2	4	1	5	2
I3	3	4	2	5
I4	5	3	4	1
I5	2	5	3	4
I6	4	2	1	5

Operational Costs of Intermediate Warehouses (Euros)

Warehouse	Operational Cost
I1	1000
I2	800
I3	1200
I4	1100
I5	700
I6	900

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IV. Mixed Integer Programming.

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LP models are very useful, but there are many situations where the planning problem in question cannot be modelled sufficiently precisely. In many problems fractionality of the variables is problematic. In a Mixed Integer Programming (MIP) model, the variables are restricted, only allowing integer values as described in the domain constraints.

As an example consider the Incredible Chairs problem again.

Example 1. The company Incredible Chairs produces two different types of chairs, A and B , where one unit (pallet of chairs) of Chair A can be sold for a profit of 4 and one unit of Chair B can be sold for a profit of 6. The chairs are produced on three production lines:

1. Line 1 requires 2 hours to produce one unit of Chair A , and it can be used at most 14 hours due to other production on the line.
2. Line 2 requires 3 hours to produce one unit of Chair B , and it can be used at most 15 hours.
3. Line 3 requires 4 hours to produce one unit of Chair A and 3 hours to produce one unit of chair B , and it can be used at most 36 hours.

How would you plan the production of the chairs?

The LP model is:

$$\begin{aligned}
 \min \quad & z = 4 \cdot x_A + 6 \cdot x_B \\
 \text{subject to} \quad & \\
 & 2 \cdot x_A \leq 14, \\
 & 3 \cdot x_B \leq 15, \\
 & 4 \cdot x_A + 3 \cdot x_B \leq 36, \\
 & x_A, x_B \geq 0.
 \end{aligned}$$

If we solve this problem we obtain: maximal profit 51.0 and the optimal values of the two decision variables, $x_A = 5.25$ and $x_B = 5.0$.

If in the above example we impose the requirement that $x_A, x_B \in \mathbb{Z}^+$, i.e. the chairs need to be produced in whole numbers, the model becomes a MIP model. The last line of the model becomes $x_A, x_B \in \mathbb{Z}^+$ instead of $x_A, x_B \geq 0$.

1 Modelling power

At first glance, the addition of integer requirements for the variables does not seem like such a big change. However, many problems are inherently integer:

- Chair production, or any type of production of integer amounts - It is hard to sell fractional chairs.
- Vehicle routing, e.g. trucks, ships, trains - a fractional ship cannot sail.
- Manpower planning - planning with fractional workers is not a good option.
- Timetabling - the classes could be divided up, but what about the teacher?

It turns out that there are many cases where one or more variables needs to be integer ($x \in \mathbb{Z}$) or binary ($x \in \{0, 1\}$).

In general, just rounding fractional variables is not guaranteed to reach an optimal solution, perhaps not even a feasible solution.

Generating a good integer solution from the Incredible chairs model is not complicated. The optimal value for Incredible chairs is 51 with the values $x_A = 5.25$ and $x_B = 5$. Because all the coefficients of x_A are positive in the constraints and all the constraints are less-equal $\leq$, x_A can simply be rounded down to $x_A = 5$, achieving an integer solution. This reduce the objective value to 50. But is this solution optimal?

In general, just rounding fractional variables is not guaranteed to reach an optimal solution, perhaps not even a feasible solution. In the Incredible Chairs case, the integer solution objective went from 51 to 50. In other cases, the changes in the objective can be far greater: Suppose a container ship has to be routed to pick up containers. If 3 containers have to be picked up from a port by a container ship with a capacity of 1000, an LP model will assign a $\frac{3}{1000}$ ship to do the job. If what is minimized is cost, the objective can increase substantially by increasing the value from $\frac{3}{1000}$ to 1.

An important special case of integer variables is binary variables i.e. $x \in \{0, 1\}$. This can model yes/no decisions, usually interpreting 1 as yes and 0 as no.

1.1 Integer modelling tricks

There are a number of standard modelling tricks which can be utilized when modelling with integer or binary decision variables. Here we briefly mention the most important constructions, and point to the assignments where they are needed.

Fixed charge link: Often a resource like a ship, depot, truck or plant, is represented by a binary variable (is it needed or not) or how many do we need. The cost of using the resource is then 0 if it is not used, but even the most limited use incurs a fixed cost. This can be stated as:

$$h(x) = \begin{cases} 0 & , x = 0, \\ f + px & , x > 0. \end{cases}$$

If we assume that there is an upperbound on $x \leq B$, then this can be modelled using binary decision variables in the following way:

$$\begin{aligned}h(x) &= fy + px, \\x &\leq By, \\y &\in \{0, 1\}.\end{aligned}$$

Such a fixed charge link could, e.g., correspond to the cost of transporting x containers on a container ship: There is an additional cost, due to extra fuel needed for the extra weight of the container, but the main cost is probably due to the need for a full ship, i.e. the y cost.

If the resource can be increased in fixed amounts, then the only change needed is to change the definition of the resource variable from binary to integer. This could correspond to, e.g., a number of container ships.

Logic on binary variables Often binary variables can be used to formulate logical relations between different binary variables. Usually, a binary variable $x = 1$ corresponds to true and $x = 0$ corresponds to false. With these definitions, we can easily define the various logic operators:

- **OR:** Given two binary variables x and y , define the v binary variable as x OR y :

$$\begin{aligned}v &\geq x, \\v &\geq y, \\v &\leq x + y.\end{aligned}$$

- **AND:** Given two binary variables x and y , define the v binary variable as x AND y :

$$\begin{aligned}v &\leq x, \\v &\leq y, \\v &\geq x + y - 1.\end{aligned}$$

- **NOT:** Given one binary variable x , define the v binary variable as NOT x :

$$v = 1 - x.$$

- **Force selection:** Given a set of i items, select at least 3 of these:

$$\sum_i x_i \geq 3.$$

- **Limit selection:** Given a set of i items, select at most 4 of these:

$$\sum_i x_i \leq 4.$$

A classical mistake in modelling is to try to model logic by using an API variables (e.g. JuMP) as variables. Perhaps this is not possible, when logic in the model needs to be modelled, it can be modelled as above.

Binary multiplication: If a MIP problem can be modelled using linear models, i.e. where the variables are not multiplied, given two binary variables x and y the multiplication of these can be formulated in a linear fashion, where the binary variable z is the result: (exactly as the x AND y operator):

$$\begin{aligned}z &\leq x, \\z &\leq y, \\z &\geq x + y - 1.\end{aligned}$$

2 Examples

2.1 Incredible Chairs

Reformulate the LP model from the Incredible Chairs problem so that it becomes a MIP model in which only discrete quantities of chairs are considered.

Solution: just change the type of decision variables in the model of the Incredible Chairs problem.

Below is a solution modelled in *JuMP* and solved with *HiGHS* in the *Julia* language.

```
using JuMP
using HiGHS

IC = Model(HiGHS.Optimizer)

@variable(IC, xA >= 0, Int)
@variable(IC, xB >= 0, Int)
@objective(IC, Max, 4*xA + 6*xB)
@constraint(IC, 2*xA <= 14)
@constraint(IC, 3*xB <= 15)
@constraint(IC, 4*xA + 3*xB <= 36)

print(IC)
optimize!(IC)

println("Termination status: $(termination_status(IC))")
if termination_status(IC) == MOI.OPTIMAL
    println("Optimal objective value: $(objective_value(IC))")
    println("xA: ", value(xA))
    println("xB: ", value(xB))
else
    println("No optimal solution available")
end
```

And the result:

```
Solving report
Status          Optimal
Primal bound    50
Dual bound      50
Gap             0% (tolerance: 0.01%)
P-D integral    0
Solution status feasible
                50 (objective)
                0 (bound viol.)
                0 (int. viol.)
                0 (row viol.)
Timing          0.00
Max sub-MIP depth 0
Nodes           0
Repair LPs      0
```

```

LP iterations      0
Termination status: OPTIMAL
Optimal objective value: 50.0
xA: 5.0
xB: 5.0

```

2.2 Startup Fund

A startup fund manager decides to use MIP models in order to optimize their profit. A new investment fund is just starting up, and €100 million is available for investment. The startup fund staff have identified nine possible investment opportunities in startup companies, the core information being the required investment and the expected value increase in 3 years.

	Investment								
	1	2	3	4	5	6	7	8	9
Estimated profit factor	1.7	1.4	1.3	2.1	1.9	1.8	1.5	2.2	1.6
Capital required	17	25	19	25	28	23	29	31	18

Investment details (millions of dollars).

Given an available budget of €100 million, the objective for the fund is to select the best possible set of investments. Since these investment opportunities are in startup companies, the full amount of capital needed for an investment must be invested (and it is not possible to invest more). Furthermore, money which is not invested is considered lost.

Formulate a binary integer program that can be used to solve this problem.

Solution: The investment manager has a classical decision: What to invest in. Since it is not possible to divide the investment in smaller sizes, making an investment is modelled with a binary variable. Then the expected profit should be maximized constrained by the budget. This problem is the classical OR problem, the knapsack problem.

Sets: $i \in Investments = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ - The set of different investment possibilities.

Parameters:

- $EstProfitFactor_i$: Investment profit factor for investment i
- $CapitalRequirements_i$: Capital requirement for project i
- $Budget = 100$: The available capital

Decision variables: Invest or not in project i : $x_i \in \{0, 1\}$. $x_i = 1$ models that an investment is made and $x_i = 0$ models that an investment is not made.

Model:

$$\begin{aligned}
 \max \quad & z = \sum_i EstProfitFactor_i \cdot x_i \\
 \text{subject to} \quad & \sum_i CapitalRequirements_i \leq Budget, \forall i.
 \end{aligned}$$

Below is a solution modelled in JuMP and solved with HiGHS in the Julia language

```

using JuMP
using HiGHS

# PARAMETERS
Investments = [1 2 3 4 5 6 7 8 9]
I=length(Investments)

# Data
EstProfitFactor = [1.7 1.4 1.3 2.1 1.9 1.8 1.5 2.2 1.6]
CapitalRequirements = [17 25 19 25 28 23 29 31 18]
Budget = 100

# Model
Investment = Model(HiGHS.Optimizer)
@variable(Investment, x[1:I], Bin) # 1 if investment is made in i

# Maximize est-profit
@objective(Investment, Max,
           sum(EstProfitFactor[i]*x[i] for i=1:I) )

# budget constraint
@constraint(Investment,
           sum(CapitalRequirements[i]*x[i] for i=1:I) <= Budget)

# solve
optimize!(Investment)
# Report results
if termination_status(Investment) == MOI.OPTIMAL
    println("RESULTS:")
    println("objective = $(objective_value(Investment))")
    for i in 1:I
        println("Investment $(Investments[i]) $(round{Int8,value(x[i])})")
    end
else
    println(" No solution")
end

```

And the result:

```

Solving report
Status          Optimal
Primal bound    7.8
Dual bound      7.8
Gap             0% (tolerance: 0.01%)
P-D integral    0.00407696755567
Solution status feasible
               7.8 (objective)
               0 (bound viol.)
               0 (int. viol.)
               0 (row viol.)
Timing          0.01
Max sub-MIP depth 1

```

```

Nodes          1
Repair LPs     0
LP iterations   8
                0 (strong br.)
                7 (separation)
                0 (heuristics)

```

RESULTS:

```

objective = 7.8000000000000001
Investment 1 1
Investment 2 0
Investment 3 0
Investment 4 1
Investment 5 0
Investment 6 1
Investment 7 0
Investment 8 1
Investment 9 0

```

Additional assignment: After creating the first investment plan, the manager decides that investments in 1 and 4 are competing companies, so it is not possible to invest in both companies and the same applies for investments in 6 and 8. Furthermore, the investments in 6 and 9 can only be performed if either investment 2 or investment 3 is done, since the developed technology in these is the basis of 6 and 9.

Solution: The additional constraints can be modeled as

```

@constraint(Investment, x[1] + x[4] <= 1)
@constraint(Investment, x[6] + x[8] <= 1)
@constraint(Investment, x[6] + x[9] <= 2*(x[2] + x[3]))

```

And the result:

Solving report

```

Status          Optimal
Primal bound    7.3
Dual bound      7.3
Gap             0% (tolerance: 0.01%)
P-D integral    0.00238953303715
Solution status feasible
                7.3 (objective)
                0 (bound viol.)
                0 (int. viol.)
                0 (row viol.)

Timing          0.02
Max sub-MIP depth 1
Nodes           1
Repair LPs      0
LP iterations   14
                0 (strong br.)
                9 (separation)
                0 (heuristics)

```

RESULTS:

`objective` = 7.3000000000000001

`Investment` 1 0

`Investment` 2 1

`Investment` 3 0

`Investment` 4 1

`Investment` 5 0

`Investment` 6 0

`Investment` 7 0

`Investment` 8 1

`Investment` 9 1

2.3 Student Teacher Meetings

In Danish schools, it is customary to have meetings during the school year, often twice a year, between the student and parents and some of the teachers of the student. The meetings take place in the evening. Before the student-teacher meeting evening, the student (or parents!) make a list of the teachers they would like to meet with. When all students have submitted their teacher meeting request, a schedule needs to be made.

The teachers each use their own classroom for their meetings. The students and their parents walk around the school and meet with the teachers they wish to talk to. A number of time slots are given, e.g., a 10 minute interval, and your job is simply to plan when, i.e. in which time slot, the meetings will occur.

The meetings are private, so a teacher can only meet one family at a time. The families would like to spend as little time as necessary on the meetings. Therefore, the overall objective is to minimize the sum of the latest time slot indices assigned to the families.

The problem is surprisingly difficult, so the data sets are all rather small and open-source solvers all struggle on even the smallest instances. It is very likely that the solver will not find the optimal solution and will terminate with a solution that is not optimal, or even without a solution.

Formulate the student-teacher meeting planning problem as a MIP, minimizing the sum of late time slots.

Solution The student teacher meeting is a scheduling problem, i.e. a problem where events are scheduled to take place at a specific place at a specific time. This problem is actually rather hard to solve, and the problem really shows the difference between the open-source solvers and commercial solvers.

Sets:

- Students: $s \in Students = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$
- Teachers: $t \in Teachers = \{1, 2, 3, 4, 5, 6\}$
- Timeslots: $ts \in TimeSlots = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$

Parameters: $StudentTeacherMeetings_{s,t}$: Incidence matrix, where

$StudentTeacherMeetings_{s,t} = 1$ if student s has to have a meeting with teacher t , otherwise 0.

Decision variables:

- Should student s meet teacher t in timeslot ts : $x_{t,ts}^s \in \{0, 1\}$

- The latest meeting for student $s : y_s \geq 0$.

Model:

$$\begin{aligned} \min \quad & z = \sum_s y_s \\ \text{subject to} \quad & \sum_{ts} x_{t,ts}^s = \text{StudentTeacherMeetings}_{s,t}, \forall s, t, \\ & \sum_t x_{t,ts}^s = 1, \forall s, ts \text{ (each student can only meet one teacher at a time),} \\ & \sum_s x_{t,ts}^s = 1, \forall t, ts \text{ (each teacher can only meet one student at a time),} \\ & ts \cdot \sum_t x_{t,ts}^s \leq y_s, \forall s, ts \text{ (force the value of } y_s \text{ to take the value of the last timeslot } ts\text{).} \end{aligned}$$

Below is a solution modelled in *JuMP* and solved with *HiGHS* in the *Julia* language

```
using JuMP
using HiGHS
# Data
include("s_04_data_15_6.jl")

# S: no of students      : in data file
# T: no of teachers      : in data file
# TS: no of timeslots    : in data
total_no_meetings=sum(StudentTeacherMeetings[:,:])
println("Total no. of meetings: $(total_no_meetings)")

# Model
stmeet = Model(HiGHS.Optimizer)

# 1 if student s has a meeting with teacher t in time ts
@variable(stmeet, x[1:S,1:T,1:TS], Bin)

for s=1:S
    for t=1:T
        if StudentTeacherMeetings[s,t]==0
            for ts=1:TS
                fix(x[s,t,ts],0; force = true)
            end
        end
    end
end

# last meeting of student s
@variable(stmeet, y[1:S] >= 0)

# Minimize summed Inconvenience
@objective(stmeet, Min, sum( y[s] for s=1:S ) )

# meetings has to occur (or not occur)
```

```

@constraint(stmeet, [s=1:S,t=1:T], sum( x[s,t,ts] for ts=1:TS ) == StudentTeacherMeetings[s,

# only one meeting pr timeslot for each student s
@constraint(stmeet, [s=1:S,ts=1:TS], sum( x[s,t,ts] for t=1:T ) <= 1)

# only one meeting pr timeslot for each teacher t
@constraint(stmeet, [t=1:T,ts=1:TS], sum( x[s,t,ts] for s=1:S ) <= 1)

# force value of y to last meeting for each student
@constraint(stmeet, [s=1:S,ts=1:TS], ts*sum( x[s,t,ts] for t=1:T ) <= y[s])

# solve
optimize!(stmeet)
println("-----");
if termination_status(stmeet) == MOI.OPTIMAL
    println("RESULTS:")
    println("objective = $(objective_value(stmeet))")
    println("Minimal num of meetings: $(sum(StudentTeacherMeetings[:,:]))")
    println("Solve time: $(solve_time(stmeet))")
else
    println(" No solution")
end

```

The data file contains the matrix and inputs.

```

StudentTeacherMeetings=[
 0 1 0 0 1 1;
 0 0 1 0 1 1;
 1 0 0 1 0 0;
 0 1 1 0 0 0;
 1 0 0 1 0 0;
 1 1 0 1 0 0;
 1 1 1 0 0 0;
 0 0 0 0 1 1;
 0 0 0 1 1 1;
 0 0 0 1 1 1;
 0 0 1 0 1 1;
 0 1 0 0 1 1;
 0 1 1 0 0 0;
 0 1 1 0 0 0;
 1 0 0 1 1 0
]
S=15
T=6
TS=11

```

And the solution:

```

Students: 15 Teachers: 6 Timeslots: 11
Total no. of meetings: 39
...
Solving report

```

```

Status                Optimal
Primal bound          66
Dual bound            66
Gap                   0% (tolerance: 0.01%)
P-D integral          2.98462585579
Solution status       feasible
                      66 (objective)
                      0 (bound viol.)
                      3.99680288865e-15 (int. viol.)
                      0 (row viol.)
Timing                28.06
Max sub-MIP depth     4
Nodes                 4615
Repair LPs            0
LP iterations          477562
                      142513 (strong br.)
                      35754 (separation)
                      27198 (heuristics)
Termination status:  OPTIMAL
-----
RESULTS:
objective = 66.0
Minimal num of meetings: 39
Solve time: 28.066952082997886
-----

```

3 Problem set

1. Work-plan for Teaching Assistants A Mathematical Programming crash course, over 9 days, is taught by 4 teaching assistants (TAs) and 2 professors. The professors give the lectures and the TAs are available for help with the exercises. Suppose that four TAs (AL, FR, JE, and MI) are the TAs in the course. In this assignment, you have to plan when they should work during the course. Each TA must work exactly 52 hours (a total of 208 hours are available for the four TAs). The professors have attempted to come up with a plan of how many TAs to assign for exercise help in each of the 8 time slots from 9 to 17. This TA demand is given below, [1](#).

To make the work easier for the TAs, we wish to take into account their personal preferences. Each TA has specified inconvenience scores for each day d and time period t in the matrix of inconvenience scores $Inconvenience_{t,d}^{ta}$ as given in the text file `s_04_inconvenience.txt`.

Formulate a mathematical model that can be used to find the optimal work plan for the TAs.

When you inspect the generated plan, you observe that the TAs work several times on each day. This is rather inconvenient. You decide to modify your model and include the restrictions that each TA can only work consecutive hours and that if a TA works on a particular day, then the TA has to work at least two hours.

Revise the mathematical model with additional restrictions.

2. Printing Precision Parts You work for a company that manufactures high quality polymer parts for OrgoCorp, a galactic bioengineering company. The quality requirements are so high that you need to use very sophisticated 3D printing machines. You need to develop a mathematical

Figure 1: TA demand for the first 9 days.

	Day								
	1	2	3	4	5	6	7	8	9
9-10	4	4	4	0	3	0	3	0	2
10-11	4	4	4	2	4	2	4	2	2
11-12	4	4	4	4	4	4	4	4	2
12-13	4	3	3	3	3	4	3	3	1
13-14	4	3	3	3	3	4	3	3	1
14-15	4	3	3	3	3	4	3	3	1
15-16	3	3	2	3	3	4	3	3	1
16-17	3	3	2	2	2	3	2	2	1

programming model to organize monthly production considering:

- *Fixed operational costs* (euros): Each 3D printing machine m has a different fixed operational costs Fm that needs to be considered when a printing machine is used to print any units.
- *Unitary production costs* (euros): Each 3D printing machine m produces parts of product type p at a different unitary costs C_{mp} .
- *Capacity* (minutes): Each 3D printing machine m has a different capacity S_m (minutes).
- *Speed* (minutes/part): Each 3D printing machine m prints a unit of product type p at a different unitary speed.
- *Demand* (units): OrgoCorp has confirmed a demand for d_p units of every product type p .

Build a model that takes into account these requirements, identifying indices, decision variables, objective function and constraints.

The CEO of OrgoCorp, The High Evolutionary, would like to set up a new contract allowing more flexibility in the delivery of units, such that:

- *Demand per period* (units): Now you need to consider that the demand for every product type p is not constant and depends on the planning period t , D_{pt} .
- *Delayed demand* (units): You may delay the production for one period. For every unit of product type p that is delayed, OrgoCorp will charge an extra delayed demand cost of B_{pt} .

Modify the model to take into account this new contract.

3. Fleet Assignment In the distant future, tensions between Earth, Mars, and the Belt are at an all-time high. The United Nations Navy (UNN) must deploy vessels to ensure the stability in earth's colony and provide security to supply chain operations across the solar system.

More specifically, Earth needs to assign five available warships to secure five colonies in need of military assistance: Ceres, Ganymede, Europa, Io, and Callisto.

Each vessel has an associated cost for each mission, that depends on factors such as suitability, operational range, and efficiency. The costs are provided in the table below:

Requirements:

Vessel / Colony	Ceres	Ganymede	Io	Europa	Callisto
UNN Agatha King	12	18	25	30	14
UNN Nathan Hale	20	14	22	24	18
UNN Donnager	30	12	18	16	10
UNN Rocinante	15	22	28	12	20
UNN Behemoth	18	20	30	14	12

1. Formulate an Integer Programming model to minimize the total cost of vessel assignments while ensuring every colony gets a vessel.
2. Draw a graph and solve the problem using graph algorithms (as a transportation model).
3. Describe a strategy to design a greedy algorithm to solve the problem.

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V. Decision theory, game theory.

Consider the following problems, for some of them you can find a solution posted on teams.

Problem set

1. You are the trainer of a basketball team playing an inter-university tournament over a weekend. You won the first game of the tournament yesterday, and now you have to decide whether to play with the same starting 5 or make changes in the team. You want to assess your decision considering the possibilities of winning (p_w) or losing (p_l) the game and of having any player injured (p_i) or all healthy (p_h) after the game. Since it is a friendly tournament, your priority is that none of your classmates ends up injured, so you establish the following priorities between the utility values of the possible outcomes: v_1 (victory – no injuries) $> v_2$ (lose – no injuries) $> v_3$ (win – any injured) $> v_4$ (lose – any injured).

- Draw a decision tree indicating the probabilities and utility values of the possible outcomes, considering that the probability of winning or losing and the probability of having any player injured or all healthy are independent and also independent of your decision to make changes in the team.
- How would the decision tree change if you consider that the probability of having any player injured is not independent of your decision to make changes?

2. Ash is a renowned Pokémon trainer. He thinks that his most beloved Pokémon, Pikachu, could have been injured in the last battle of the Winners Trophy play-offs. Therefore, he asks himself: *should he use Pikachu in the next battle against Team Rocket?*

The trainer thinks that there is a probability of 0.1 of victory. However, if Pikachu participates in the fight, his injury could worsen. Ash thinks that there is a probability of 0.2 that Pikachu is actually injured from the last battle.

The utilities (in points) that the trainer employs are:

- if Pikachu fights, and Ash wins the battle, and Pikachu is not injured, +100;
- if Pikachu fights, and Ash wins the battle, and Pikachu is injured, +50;
- if Pikachu fights, and Ash loses the battle, and Pikachu is not injured, 0;
- if Pikachu fights, and Ash loses the battle, and Pikachu is injured, -50;
- If Pikachu does not fight and Pikachu is injured, the utility is -10;

- If Pikachu does not fight and Pikachu is not injured, the utility is 0.

Ash wants to use decision theory to support his decision-making in the battle against Team Rocket.

- a. Draw the decision tree for this problem considering that the probabilities of the result of the battle and the injury are independent.
- b. Determine which is the best decision for Ash.
- c. Ash could acquire further information about the probabilities of the result of the battle. The additional information determines the result of the battle (win or lose), and based on previous results, it has a certain accuracy to successfully predict the result of the battle. Draw again the decision tree to incorporate into the decision problem the acquisition of this imperfect information about the result of the battle, indicating in the corresponding branches the revised probabilities

3. Alice and Bob are two classmates that want to celebrate the end of the course with a game. They will go to their favourite bar and Bob will randomly select a bottle of wine from the wine list. Alice will do a blind testing of the wine, with no possibility to see the wine selected by Bob. If Alice guesses the type of wine (red or white) just by holding her glass (by smell), Bob will pay 20€. Otherwise, if she fails, Alice will pay Bob 10€. She may also taste the wine for 5€ (that is, Bob will discount 5€ of the bet if she guesses the type of wine, or Alice will pay 5€ extra if she does not).

Alice thinks she can use decision theory to improve her chances of winning the bet. She knows that in the wine list of their favourite bar, there are 15 bottles of red wine and 11 of white wine. She has also done some blind wine tasting to prepare for the bet and found out that it is actually not that easy to guess the colour of the wine just by tasting it blindfolded. She guessed correctly $\frac{3}{4}$ of the times she tried a white wine and $\frac{2}{4}$ of the times she tried a red wine.

- a. Draw the decision tree for Alice without taking into account that she may test the wine to get additional information.
- b. Modify the decision tree for Alice now taking into account that she may test the wine.
- c. What is the best alternative for Alice?

4. Your company, *Altered Silicon*, manufactures solar energy panels, and it has received an urgent order for 40 panels from a new German customer which is willing to pay 1000€ per panel. Altered Silicon has committed its production capacity to other orders, and the only alternative left to serve the new order is to subcontract the production involved.

Altered Silicon has located two possible suppliers:

- Supplier A: based in Romania, placed an offer of 500€ per panel
- Supplier B: from Poland, which would sell each panel for 550€

The cost to transport one batch of 40 panels from either of the two suppliers to the German firm is 1000€. Nonetheless, the customer's technical specifications are very demanding, so Altered Silicon estimates that Supplier A could supply one correct batch with a probability of 70%, whereas Supplier B could do it with a probability of 80%. If the batch were incorrect, Altered Silicon could sell it to another customer at the price of 200€ per panel. Altered Silicon is considering the possibility of asking for a test panel from each supplier to test it before placing an order. Altered

Silicon's testing system guarantees that a faulty panel is always detected, while there is a 5% margin of error with correct panels. The testing system is prepared to assess two test panels, and the overall cost (cost of the two panels + transport + cost of the test) comes to 2000€.

- a. Draw the corresponding decision tree and indicate with a letter per branch the associated probability or the associated cost (depending on the case)
- b. Solve the decision tree
- c. Up to how much should Altered Silicon pay for the test?

5. Two manufacturers, M1 and M2, compete for the sales of two different product lines which are equally profitable. In both cases, M2's sales volume triples that of M1. Due to some technological advances, both manufacturers can make substantial improvements to the two products, but they are not sure as to which development and commercialisation strategy they should adopt. If the improvements to the two products are done at the same time, neither manufacturer can have them ready to sell under 12 months. One alternative is to carry out an "intensive programme" to first develop one of the two products and to try to commercialise it before the competition does. If this were the case, M2 can have one product ready to sell in 9 months, while M1 needs 10 months (due to previous commitments in its production installations). Either manufacturer can have the second product ready after another 9-month period. For either product line, if both manufacturers commercialised the improved models simultaneously, it is estimated that M1 can increase the percentage of all the future sales of this product by 8% of the total (from 25% to 33%). Likewise, M1 can increase its total sales in 20, 30 and 40% of the total if it commercialises the product 2, 6 and 8 months before M2 does, respectively. Moreover, M1 can lose 4, 10, 12 and 14% of the total if M2 manages to commercialise it 1, 3, 7 and 10 month(s) before it, respectively.

Formulate this problem as if it were a zero-sum game with two players and then establish the strategy the manufacturers must adopt according to the MinMax criterion.

6. Two students, Alice and Bob, are playing a game named Bluff, that goes as follows:

- Alice writes a number on a piece of paper. Without showing Bob what she has written, she tells him what she wrote. Alice might be lying or telling the truth.
- Then Bob must guess if Alice is lying or telling the truth, or he can choose to pass and not to continue playing.
- If Bob discovers that Alice lied, Alice must pay Bob 50€. If Bob wrongly accuses Alice of lying, Bob pays her 25€. If Alice told the truth and Bob guessed this, then Alice pays Bob 5€. If Alice lied and Bob does not guess that she lied, then Bob pays Alice 24€.
- Should Bob decide to pass, then neither player wins or loses anything.

- a. Consider this a 2-player's zero-sum game and write down the tabular representation.
- b. How would you determine the game mean value and the mixed strategies of both players in this case?
- c. Using a method to calculate the mixed strategies, a solver provides the following solution for Bob ($u_1 = 0, u_2 = 0, u_3 = 0.02$), where u_1 is the decision variable associated to the alternative "Alice lied", u_2 the decision variable associated to the alternative "Alice told the truth" and u_3 the decision variable associated to "Pass". Write down the model and determine his optimal strategy.

7. Suppose there are two political parties, party A and B (with opposite views). They are competing in an election. Each party has three possible strategies: they can focus their campaign on young citizens (below 30), middle-aged citizens (30 – 50), or senior citizens (over 50). The parties use national statistics to estimate the number of votes that can shift in each demographic group. Here are the estimated values for each demographic group:

- Young voters: They represent 60% of the undecided voters.
- Middle-aged voters: They represent 25% of the undecided voters.
- Senior citizens: They represent 15% of the undecided voters.

From past elections, party A has collected the following estimation about undecided voters on each scenario:

- Young voters: If both parties focus on young voters, party A will get 40% of the young voters and party B 60% of the young voters. If party A focuses on the young voters and party B does not, party A will get 70% of the young voters. If party B focuses on the young voters and party A does not, party A will get 10% of the young voters. If neither party focuses on the young voters, party A will get 40% of the young voters.
- Middle-aged voters: If both parties focus on middle-aged voters, party A will get 60% of the middle-aged voters and party B 40% of the middle-aged voters. If party A focuses on the middle-aged voters and party B does not, party A will get 90% of the middle-aged voters. If party B focuses on the middle-aged voters and party A does not, party A will get 30% of the middle-aged voters. If neither party focuses on the middle-aged voters, party A will get 60% of the middle-aged voters.
- Senior citizens: If both parties focus on senior citizens, party A will get 70% of the senior citizens and party B 30% of the senior citizens. If party A focuses on the senior citizens and party B does not, party A will get 90% of the senior citizens. If party B focuses on the senior citizens and party A does not, party A will get 20% of the senior citizens. If neither party focuses on the senior citizens, party A will get 70% of the senior citizens.

Model the strategies of both political parties, following these steps:

1. Identify the alternatives for both parties and draw a tabular representation of the game were Player A is party A.
2. Complete the tabular representation of the game, estimating the total percentage of undecided voters that would vote for each party.
3. Calculate the strategy of each party using the MinMax criteria.

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VI. Problems.

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1 Linear Programming

1.1 Renting Warehouses

Problem definition A firm has realised that it will not have sufficient storage space for the next three months. Details of the additional storage requirements for the next 3 months are provided in Table 1.

Table 1: Additional storage space requirements.

Month	January	February	March
Space required (1,000 m ²)	25	10	20

To cover its requirements, the firm plans to rent additional space on a short-term basis. At the beginning of each month, the firm can rent any amount of space for any number of months. It can pay separate rents of different quantities of space and/or over different periods of time. For instance, during the first month it could rent 20000 m² for 2 months, and rent 5000 m² separately

for 1 month. It could also acquire new rented spaces before the previous ones expire. The costs per 1000 m^2 of rented space in accordance with duration are provided in Table 2

Table 2: Rent costs.

Rent duration	1 month	2 months	3 months
Cost (\$ per 1,000 m^2)	280	450	600

Build a linear programming model whose solution provides a rent policy that covers space requirements at a minimum cost.

Solution

Decision variables:

- X_{11} = Quantity of space (m^2) the firm rents in month 1 for month 1.
- X_{12} = Quantity of space (m^2) the firm rents in month 1 for month 2.
- X_{13} = Quantity of space (m^2) the firm rents in month 1 for month 3.
- X_{21} = Quantity of space (m^2) the firm rents in month 2 for month 1.
- X_{22} = Quantity of space (m^2) the firm rents in month 2 for month 2.
- X_{31} = Quantity of space (m^2) the firm rents in month 3 for month 1.

Objective function:

$$\text{To minimise } z = 280(X_{11} + X_{21} + X_{31}) + 450(X_{12} + X_{22}) + 600X_{13}$$

Constraints:

$$\begin{aligned} X_{11} + X_{12} + X_{13} &\geq 25000 \\ X_{12} + X_{13} + X_{21} + X_{22} &\geq 10000 \\ X_{13} + X_{22} + X_{31} &\geq 20000 \\ X_{11}, X_{12}, X_{13}, X_{21}, X_{22}, X_{31} &\geq 0 \end{aligned}$$

The objective function minimises the cost of the space to be rented. The first three constraints ensure that the quantity of space required is rented for months 1, 2 and 3, respectively. The last constraint imposes non-negativity to all the variables.

1.2 Crop Planning

Problem definition A citrus firm in Valencia possesses 120 acres (1 acre = 4074 m^2) and it is planning to sow at least three crops. The seeds for crops A, B and C cost \$40, \$20 and \$30 per acre, respectively. It intends to invest a maximum of \$3400 in seeds. Seeds A, B and C require 1, 2 and 1 working day(s) per acre, respectively, and 170 working days available are considered. If the owner can obtain a profit of \$100 per acre with seed A, \$300 per acre with seed B and \$200 per acre with seed C, how many acres should the owner sow of each seed to maximise the profit?

1. Model the programme like a linear programming model.
2. Is it possible to sow 40 acres with seed A, 50 acres with seed B and 30 acres with seed C?
3. Is it possible to not sow seed A, and to sow 40 acres with seed B and 70 acres with seed C?
4. What is the optimum solution? What is the optimum objective function value?
5. What capacity is completely used with the optimum combination?
6. Where are the idle capacities?

7. If the budget was increased by \$1, how much would the total profit increase by? Why?
8. If the working days available increased by one extra day, how much would the total profit increase by? Why?
9. If the initial budget was changed to \$3401, what would happen to the optimum objective function value?
10. If the budget was limited to \$800 for seed B, what would happen to the optimum objective function value?
11. If the profit per acre with seed B was \$400, would the optimum amounts in acres be the same? Why?
12. If the profit per acre with seed C were \$310, would the optimum amounts in acres be the same? Why?

Solution

Table 3: Solution obtained for problem 2.

Decision variable	Solution value	Unit cost or profit c(j)	Total contribution	Reduced cost	Basic status	Allowable Min. c(j)	Allowable Max. c(j)
X_1	0	100.0000	0	-100.0000	At bound	-M	200.0000
X_2	50.0000	300.0000	15000.0000	0	Basic	200.0000	400.0000
X_3	70.0000	200.0000	14000.0000	0	Basic	150.0000	300.0000
Objective Function	Max=	29000.0000					
Constraint	Left-hand Side	Direction	Right-hand side	Slack or Surplus	Shadow price	Allowable Min. RHS.	Allowable Max. RHS.
C1	120.0000	<=	120.0000	0	100.0000	85.0000	127.5000
C2	3100.0000	<=	3400.0000	300.0000	0	3100.0000	M
C3	170.0000	<=	170.0000	0	100.0000	140.0000	240.0000

1. Decision variables:

- X_1 = number of acres to sow with seed A.
- X_2 = number of acres to sow with seed B.
- X_3 = number of acres to sow with seed C.

Objective function (maximise the profit):

$$\text{Max } z = 100X_1 + 300X_2 + 200X_3$$

Constraints:

$$\begin{aligned}
 X_1 + X_2 + X_3 &\leq 120 && \text{(total number of acres available)} \\
 40X_1 + 20X_2 + 30X_3 &\leq 3400 && \text{(budget available for seeds)} \\
 X_1 + 2X_2 + X_3 &\leq 170 && \text{(number of working days available)} \\
 X_1 + X_2 + X_3 &\geq 0
 \end{aligned}$$

The solution to the proposed problem has been obtained with a commercial solver (see the sample in Table 3).

2. It would not be the optimum solution.

3. It would be the optimum solution.
4. Not sowing seed A, sowing 40 acres with seed B and 70 acres with seed C to obtain a profit of \$29000.
5. The total number of acres available and the total number of working days.
6. In the available budget for seeds.
7. It would increase by nothing because it is an idle capacity.
8. By \$100, which corresponds to the shadow price of the corresponding constraint.
9. Nothing because there is already a surplus of \$300.
10. It would be necessary to add a new constraint to the initial model, $20X_2 \leq 800$, and the problem would have to be solved again in order to answer this question.
11. All we can state with any certainty is that the optimum solution structure would be the same. This is because 400 is within the limit of the ranges for the profit coefficient of X_2 .
12. All we can state with any certainty is that the optimum solution structure would be the same. This is because 310 is within the limit of the ranges for the profit coefficient of X_3 .

2 Integer programming

2.1 Distribution of Air-Conditioning Units

Problem definition A North American air-conditioning firm has production plants in Portland and Flint. It has to supply a certain number of units in its distribution centres located in Los Angeles and Atlanta. The delivery costs are summarised in Table 4. The supply and demand data

Table 4: Delivery costs between distribution centres.

Production plants	Distribution centres	Delivery costs
Portland	Los Angeles	\$30
	Atlanta	\$40
Flint	Los Angeles	\$60
	Atlanta	\$50

are provided in Table 5 as number of units:

Table 5: Supply and demand data.

Supply	Available units	Demand	Required units
Portland	200	Los Angeles	300
Flint	600	Atlanta	400

1. Consider an integer linear programming model that determines how the deliveries from Portland and Flint should be carried out in order to minimise delivery costs.
2. Solve the problem and interpret the results.

Solution

a. Decision variables:

- X_1 = Units delivered from Portland to Los Angeles.

- X_2 = Units delivered from Portland to Atlanta.
- X_3 = Units delivered from Flint to Los Angeles.
- X_4 = Units delivered from Flint to Atlanta.

Objective function:

$$\text{Minimize } z = 30X_1 + 40X_2 + 60X_3 + 50X_4$$

Constraints:

$$X_1 + X_2 \leq 200$$

$$X_3 + X_4 \leq 600$$

$$X_1 + X_3 \geq 300$$

$$X_2 + X_4 \geq 400$$

$$X_1, X_2, X_3, X_4 \geq 0 \text{ and integer values}$$

b. Step 1: Standardising the problem

A minimisation problem is dealt with by multiplying the objective function by -1 and we transform it into a maximisation problem.

$$\text{Min } z = \text{Max } w = -30X_1 - 40X_2 - 60X_3 - 50X_4 + 0S_1 + 0S_2 - 0S_3 - 0S_4 - MA_3 - MA_4$$

$$X_1 + X_2 + S_1 = 200$$

$$X_3 + X_4 + S_2 = 600$$

$$X_1 + X_3 - S_3 + A_3 = 300$$

$$X_2 + X_4 - S_4 + A_4 = 400$$

Step 2: Generating an initial solution (Tables 6, 7, 8 and 9).

Table 6: Simplex Method.

C_j	C_j Base	-30 X_1	-40 X_2	-60 X_3	-50 X_4	0 S_1	0 S_2	0 S_3	0 S_4	-M A_3	-M A_4	R	Ratio
0	S_1	1	1	0	0	1	0	0	0	0	0	200	200
0	S_2	0	0	1	1	0	1	0	0	0	0	600	600/0
-M	A_3	1	0	1	0	0	0	-1	0	1	0	300	300
-M	A_4	0	1	0	1	0	0	0	-1	0	1	400	400/0
	$C_j - Z_j$	-30+M	-40+M	-60+M	-50+M	0	0	-M	-M	0	0		

Table 7: Simplex Method.

C_j	C_j Base	-30 X_1	-40 X_2	-60 X_3	-50 X_4	0 S_1	0 S_2	0 S_3	0 S_4	-M A_3	-M A_4	R	Ratio
-30	X_1	1	1	0	0	1	0	0	0	0	0	200	200/0
0	S_2	0	0	1	1	0	1	0	0	0	0	600	600
-M	A_3	0	-1	1	0	-1	0	-1	0	1	0	100	100/0
-M	A_4	0	1	0	1	0	0	0	-1	0	1	400	400
	$C_j - Z_j$	0	-10	-60+M	-50+M	30-M	0	-M	-M	0	0		

$$X_1 = X_2 = X_3 = X_4 = S_3 = S_4 = 0$$

The solution to the problem is:

$$\begin{aligned} X_1 &= 200 \\ X_2 &= 0 \\ X_3 &= 100 \\ X_4 &= 400 \\ z &= 32000 \end{aligned}$$

Table 8: Simplex Method.

	C_j	-30	-40	-60	-50	0	0	0	0	-M	-M	
C_j	Base	X_1	X_2	X_3	X_4	S_1	S_2	S_3	S_4	A_3	A_4	R
-30	X_1	1	1	0	0	1	0	0	0	0	0	200
0	S_2	0	-1	1	0	0	1	0	1	0	-1	200
-M	A_3	0	-1	1	0	-1	0	-1	0	1	0	100
-50	X_4	0	1	0	1	0	0	0	-1	0	1	400
	$C_j - Z_j$	0	40 - M	-60 + M	0	30 - M	0	-M	-50	0	-M + 50	

Table 9: Simplex Method.

	C_j	-30	-40	-60	-50	0	0	0	0	-M	-M	
C_j	Base	X_1	X_2	X_3	X_4	S_1	S_2	S_3	S_4	A_3	A_4	R
-30	X_1	1	1	0	0	1	0	0	0	0	0	200
0	S_2	0	0	0	0	1	1	1	1	-1	-1	200
-60	X_3	0	-1	1	0	-1	0	-1	0	1	0	100
-50	X_4	0	1	0	1	0	0	0	-1	0	1	400
	$C_j - Z_j$	0	-20	0	0	-30	0	-60	-50	60 - M	50 - M	

In other words, send 200 units from Portland to Los Angeles, 100 go from Flint to Los Angeles, and 400 leave Flint to Atlanta, with a total cost of \$32,000.

2.2 Allocation in a Lawyers' Office

Problem definition A lawyers' office has taken on five new cases, and each one can be adequately handled by any of the five more recent partners. Given the difference in experience and practice, lawyers will spend different times on their cases. One of the most experienced partners has estimated requirements in terms of time (in hours; see Table [11](#)).

Table 10: Estimation of the time required per case and lawyer.

	Case 1	Case 2	Case 3	Case 4	Case 5
Lawyer 1	45	22	30	65	15
Lawyer 2	180	163	85	148	68
Lawyer 3	21	17	193	59	65
Lawyer 4	18	83	16	70	115
Lawyer 5	97	75	20	70	71

In order to determine the optimum way of allocating cases to lawyers so that each one takes on a different case and that the total time of hours spent is the minimum formulate this problem as an integer linear programming model.

Solution

Decision variables:

This problem can be modelled as an allocation problem. As it is a matter of deciding which lawyer takes on each case, we define X_{ij} which takes a value of 1 if lawyer i is allocated case j , and 0 otherwise, where $i, j = 1, \dots, 5$. These 25 variables are the decision variables of the problem.

Objective function:

Our objective is to optimally allocate lawyers to cases (or vice versa). Evidently, one allocation is better than another if the total number of hours spent on preparing cases is lower. Hence, the objective function must minimise the number of hours that the five lawyers spend on preparing the five cases.

$$\begin{aligned} \text{Minimize } z = & 45X_{11} + 22X_{12} + 30X_{13} + 65X_{14} + 15X_{15} + 180X_{21} + 163X_{22} \\ & + 85X_{23} + 148X_{24} + 68X_{25} + 21X_{31} + 17X_{32} + 193X_{33} + 59X_{34} \\ & + 65X_{35} + 18X_{41} + 83X_{42} + 16X_{43} + 70X_{44} + 115X_{45} + 97X_{51} \\ & + 75X_{52} + 20X_{53} + 70X_{54} + 71X_{55} \end{aligned}$$

Constraints:

The constraints of this problem emerge because each lawyer is expected to take on a different case. If lawyer 1 can take on only one case, then one, and only one, of variables $X_{1j}, j = 1, \dots, 5$ takes a value of 1, while the other four should take a value of zero:

$$\begin{aligned} X_{11} + X_{12} + X_{13} + X_{14} + X_{15} &= 1 \text{ (1 case for lawyer 1)} \\ X_{21} + X_{22} + X_{23} + X_{24} + X_{25} &= 1 \text{ (1 case for lawyer 2)} \\ X_{31} + X_{32} + X_{33} + X_{34} + X_{35} &= 1 \text{ (1 case for lawyer 3)} \\ X_{41} + X_{42} + X_{43} + X_{44} + X_{45} &= 1 \text{ (1 case for lawyer 4)} \\ X_{51} + X_{52} + X_{53} + X_{54} + X_{55} &= 1 \text{ (1 case for lawyer 5)} \end{aligned}$$

Besides, together with the fact that there are as many lawyers as cases, each case can be prepared by only one lawyer:

$$\begin{aligned} X_{11} + X_{21} + X_{31} + X_{41} + X_{51} &= 1 \text{ (1 lawyer for case 1)} \\ X_{12} + X_{22} + X_{32} + X_{42} + X_{52} &= 1 \text{ (1 lawyer for case 2)} \\ X_{13} + X_{23} + X_{33} + X_{43} + X_{53} &= 1 \text{ (1 lawyer for case 3)} \\ X_{14} + X_{24} + X_{34} + X_{44} + X_{54} &= 1 \text{ (1 lawyer for case 4)} \\ X_{15} + X_{25} + X_{35} + X_{45} + X_{55} &= 1 \text{ (1 lawyer for case 5)} \\ X_{ij} &\in \{0, 1\}, i, j = 1 \dots 5 \end{aligned}$$

3 Non-Linear programming

3.1 Placing a Handrail

Problem definition The intention is to place a handrail on a terrace with an empty space in its centre in such a way that the handrail goes around the interior and exterior terrace perimeters. This empty space, which is rectangular in shape, measures 10 m wide by 18 m long (see Figure 1). Between the interior and the exterior terrace perimeters, a walkway must be left, which is the same width on all sides. If the handrail measures 250m:

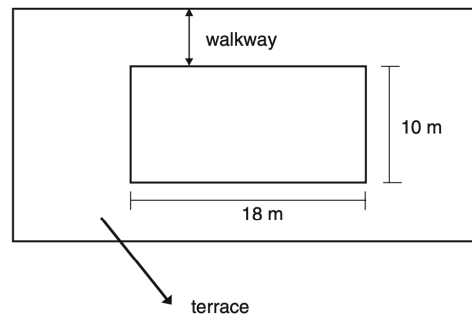


Figure 1: Sketch of the terrace.

1. Formulate a non-linear programming model that maximises the area occupied by the walkway to improve circulation in it.
2. Apply the Kuhn-Tucker conditions to the model in the previous section and indicate the metres that the outer terrace sides and the walkway width measure.
3. This problem could have been formulated by linear programming and by maximising the walkway width in this case, but how?

Solution

a. Decision variables:

-
- X_1 metres that the outer terrace side measures in parallel with the 10-m side of the empty space.
- X_2 metres that the outer terrace side measures in parallel with the 18-m side of the empty space.

Objective function (maximize the walkway area):

$$\text{Max } f = X_1 X_2 - 180 \text{ rectangular outer area} - \text{empty space}$$

Constraints:

$$\begin{aligned} (X_1 - 10)/2 &= (X_2 - 18)/2 && \text{(The walkway width is equal on all sides)} \\ 2(X_1 - 10)(X_2 - 18) &\leq 250 && \text{(Only a 250-m handrail is available)} \\ X_1, X_2 &\geq 0 \end{aligned}$$

b. The canonical form of the problem:

$$\begin{aligned} \text{Max } f &= X_1 X_2 - 180 \\ 0.5X_1 - 0.5X_2 + 4 &\leq 0 \\ 0.5X_1 + 0.5X_2 - 4 &\leq 0 \\ 2X_1 + 2X_2 - 194 &\leq 0 \end{aligned}$$

The Lagrangian function:

$$\begin{aligned} L &= X_1 X_2 - 180 + \lambda_1(0.5X_1 - 0.5X_2 + 4) + \lambda_2(-0.5X_1 + 0.5X_2 - 4) \\ &\quad + \lambda_3(2X_1 + 2X_2 - 194) \end{aligned}$$

Kuhn-Tucker conditions:

Gradient condition:

$$\begin{aligned} \frac{\partial L}{\partial X_1} &= X_2 + 0.5\lambda_1 - 0.5\lambda_2 + 2\lambda_3 = 0 \\ \frac{\partial L}{\partial X_2} &= X_1 - 0.5\lambda_1 + 0.5\lambda_2 + 2\lambda_3 = 0 \end{aligned}$$

Feasibility condition:

$$\begin{aligned} 5X_1 - 0.5X_2 + 4 &\leq 0 \\ -0.5X_1 + 0.5X_2 - 4 &\leq 0 \\ 2X_1 + 2X_2 - 194 &\leq 0 \end{aligned}$$

Orthogonality condition:

$$\begin{aligned} \lambda_1(0.5X_1 - 0.5X_2 + 4) &= 0 \\ \lambda_2(-0.5X_1 + 0.5X_2 - 4) &= 0 \\ \lambda_3(2X_1 + 2X_2 - 194) &= 0 \end{aligned}$$

Non-positivity condition

$$\lambda_1, \lambda_2, \lambda_3 \leq 0$$

Solution:

If $\lambda_1 = \lambda_2 = \lambda_3 = 0 \rightarrow X_1 = X_2 = 0$ (not possible)
 If $\lambda_1 = \lambda_2 = 0$ and $\lambda_3 \neq 0 \rightarrow X_1 = X_2$ (not possible)
 If $\lambda_1 = \lambda_3 = 0$ and $\lambda_2 \neq 0 \rightarrow X_1 = -X_2$ (not possible)
 If $\lambda_1 = \lambda_2 \neq 0$ and $\lambda_3 = 0 \rightarrow X_1 = -X_2$ (not possible)
 If $\lambda_1 = \lambda_2 = \lambda_3 \neq 0$:
 $(0.5X_1 - 0.5X_2 + 4) = 0$
 $(-0.5X_1 + 0.5X_2 - 4) = 0$
 $(2X_1 + 2X_2 - 194) = 0.$

There are three equations with two unknown quantities, where $X_1 = 44.5$, $X_2 = 52.5$ and the walkway width is 17.25 m.

The Hessian Matrix of $f(x_1, x_2) = (0, 1, 1, 0)$. Thus f is semi-defined negative, the function is concave and, therefore, the values obtained are a global maximum of f .

c. Decision variables:

- X metres of walkway width.
- X_1 metres that the outer terrace side measures in parallel with the 10-m side of the empty space.
- X_2 metres that the outer terrace side measures in parallel with the 18-m side of the empty space.

Objective function:

$$\text{Max } f = X$$

Constraints:

$$X = (X_1 - 10)/2 \text{ (walkway width equal on all sides)}$$

$$X = (X_2 - 18)/2 \text{ (walkway width equal on all sides)}$$

$$2[(2X + 18) + (2X + 10)] + 56 \leq 250 \text{ (outer perimeter + inner perimeter smaller than or equal to 250 m)}$$

$$X, X_1, X_2 \geq 0.$$

3.2 Designing and Planning a Project to Launch a Scientific Balloon

Problem definition A research centre has to design and launch a balloon with an X-ray telescope and other scientific equipment. An approximate measure of performance can be expressed in terms of the height that the balloon reaches and the weight of the equipment raised. Evidently, height itself is given by the volume of the balloon. Based on past experience, it was concluded that to maximise a satisfactory function for performance, then:

$$P = f(V, W) = 100V - 0.3V^2 + 80W - 0.2W^2$$

where V is the volume of the balloon and W is the weight of the equipment. The project budget is limited to \$1040. The cost associated with volume V is $2V$, and the cost of the equipment is $4W$. To obtain a reasonable balance between performance owing to the height and weight of the scientific equipment, the design must fulfil constraint $80W \geq 100V$.

Obtain an optimum design in terms of the volume and weight of the equipment, and solve it using the Kuhn-Tucker algorithm.

Solution

Decision Variables:

V = volume of the balloon;
 W = weight of the equipment;

Objective Function:

$$\text{Maximise } = f(V, W) = 100V - 0.3V^2 + 80W - 0.2W^2$$

Constraints:

$$2V + 4W \leq 1040 \rightarrow 2V + 4W - 1040 \leq 0$$

$$80W \geq 100V \rightarrow 100V - 80W \leq 0$$

Non-negativity:

$$V, W \geq 0$$

Lagrangian Function:

$$L(V, W, \lambda_1, \lambda_2) = 100V - 0.3V^2 + 80W - 0.2W^2 + \lambda_1(2V + 4W - 1040) + \lambda_2(100V - 80W)$$

Gradient condition:

$$\partial L / \partial V = 100 - 0.6V + 2\lambda_1 + 100\lambda_2 = 0$$

$$\partial L / \partial W = 80 - 0.4W + 4\lambda_1 - 80\lambda_2 = 0$$

Orthogonality condition:

$$\lambda_1(2V + 4W - 1040) = 0 \quad \lambda_2(100V - 80W) = 0$$

Feasibility condition:

$$\partial L / \partial \lambda_1 = 2V + 4W - 1040 \leq 0$$

$$\partial L / \partial \lambda_2 = 100V - 80W \leq 0$$

Non-positivity condition:

$$\lambda_1, \lambda_2 \leq 0$$

By analysing the possible values of λ_1 and λ_2 , we obtain:

$$V = 148.57;$$

$$W = 185.71;$$

$$\lambda_1 = -2.57;$$

$$\lambda_2 = -0.06;$$

Based on the Hessian Matrix of f , $H = [-0.6, 0, 0, -0.4]$, it is possible to conclude that the objective function is concave. Therefore the critical point obtained is a global maximum.

4 Network Modelling

4.1 Assigning Cases in a Lawyer's Office

Problem definition A lawyer's office has accepted five new cases, and each can be suitably managed by any of the five more recent partners. Owing to the difference in experience and practice, lawyers will employ different times on their cases. One of the most experienced partners has estimated time requirements (in hours) which are indicated in Table 11.

Table 11: Estimation of the time requirements per lawyer and case.

	Case 1	Case 2	Case 3	Case 4	Case 5
Lawyer 1	45	22	30	65	15
Lawyer 2	180	163	85	148	68
Lawyer 3	21	17	193	59	65
Lawyer 4	18	83	16	70	115
Lawyer 5	97	75	20	70	71

Formulate this problem as a Graph Theory model to optimally determine the assignment of cases to lawyers so that each one is dedicated to a different case and that the total time spent in hours is minimum.

Solution

This is a maximal flow model at a minimum cost. The structure of the graph is indicated in Figure 2.

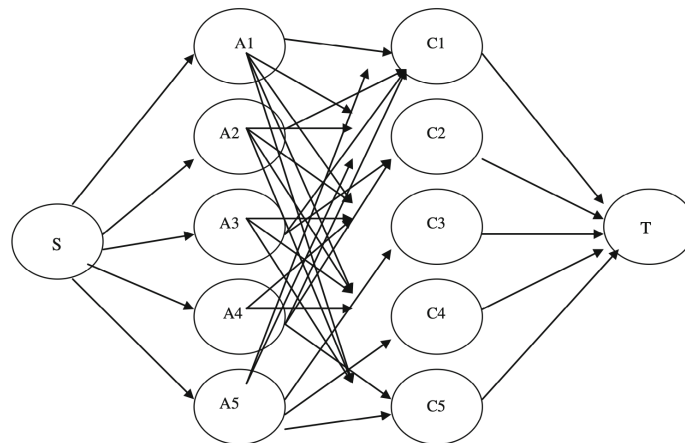


Figure 2: The assigning cases to lawyers graph.

Each arc must contain the following information:

- A_i = minimum capacity.
- B_i = maximum capacity.
- C_i = minimum cost.

In Figure 2, the arcs going from source node S to each lawyer have a minimum capacity of 0, a maximum capacity of 1 and a minimum cost of 0; that is: $(0, 1, 0)$.

The arcs going from each lawyer to each case will have a minimum capacity of 0, a maximum capacity of 1 and the cost of the corresponding hours. For example, the arc that goes from lawyer 1 to case 1, will be: $(0, 1, 145)$.

Finally, the arcs going from each case to destination node T will have a minimum capacity of 0, a maximum capacity of 1 and a minimum cost of 0; that is: $(0, 1, 0)$.